Representational Structure of Ordered Contexts

Siddhant Iyer[1,5], Rhiannon L. Cowan[2], Shervin Rahimpour[2], Stefano Fusi[3,5], Ben Shofty[2], Elliot H. Smith[2], and Nuttida Rungratsameetaweemana[4,5]

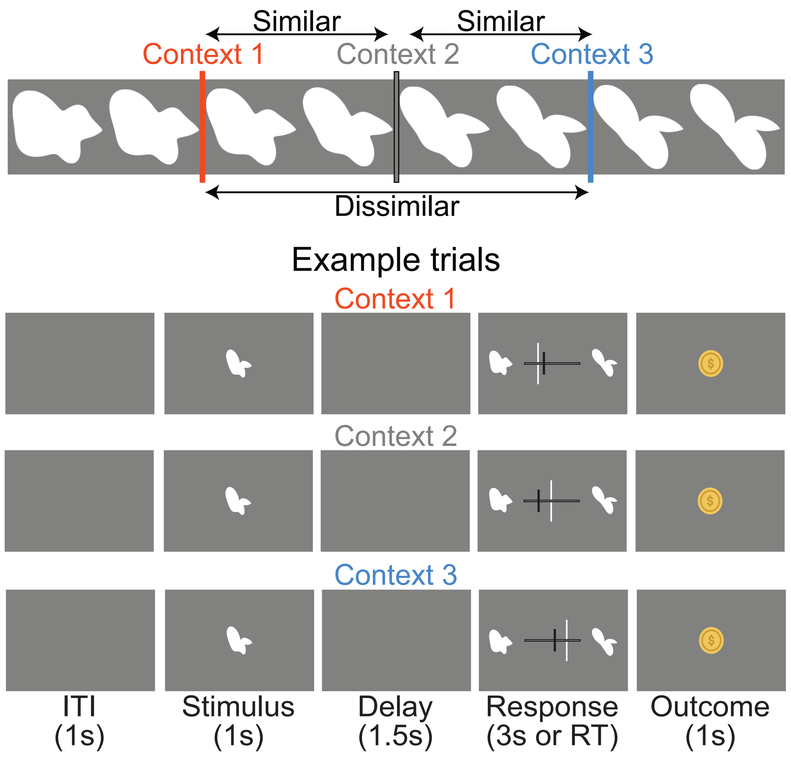
[1]Doctoral Program in Neurobiology and Behavior; [2]Department of Neurosurgery, University of Utah; [3]Department of Neuroscience; [4]Department of Biomedical Engineering; [5]Columbia University

contacts: si2442@columbia.edu, nr2869@columbia.edu

Abstract

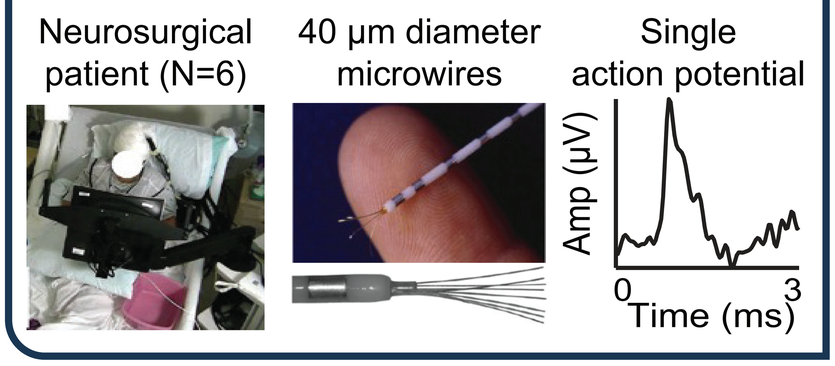
- Human behavior and their Prefrontal & Hippocampal representations can generalize across similar contexts and reconfigure between different ones.
- Can the human brain efficiently balance the retention of generalizable features, while reconfiguring those that change?
- By varying contexts along a linear space, we ask if the dissimilarity between contexts modulates the degree of reconfiguration of generalizable features.
- Human behavior & network solutions support our hypothesis, scaffolding future analyses of PFC/HPC neurons from neurosurgical patients.

Contextual classification task

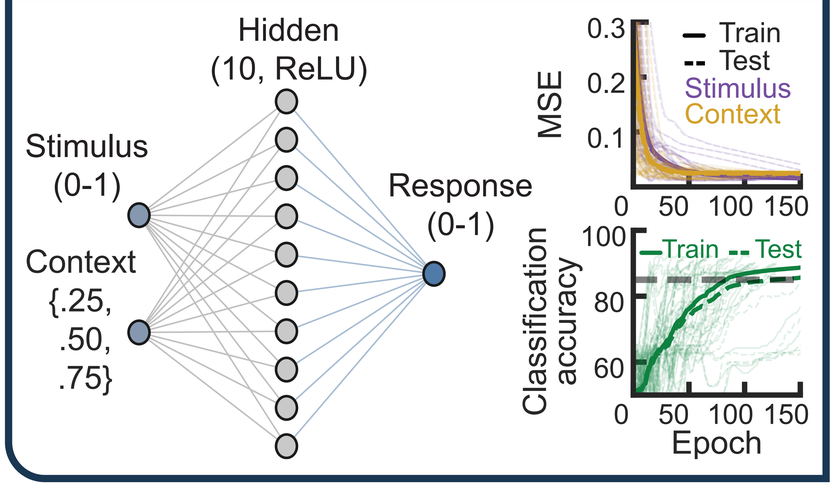


Session: 2 blocks per context, 240 trials

Neural recordings

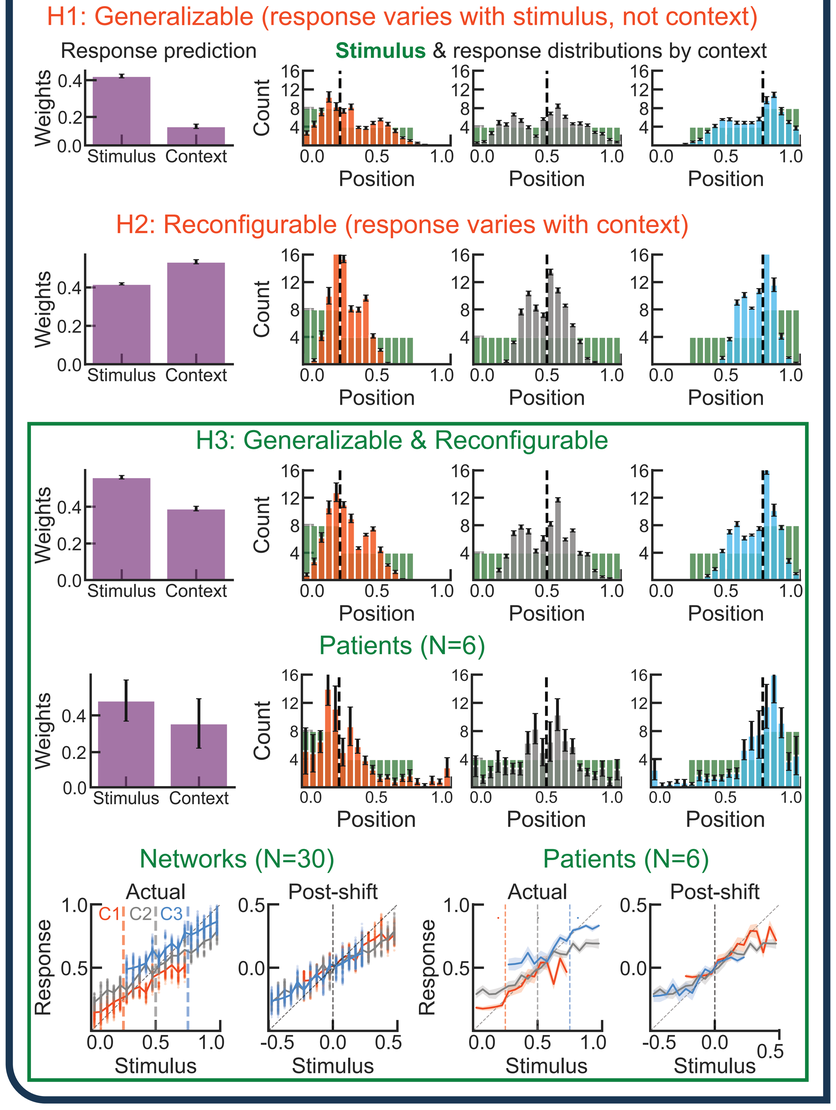


Network simulations



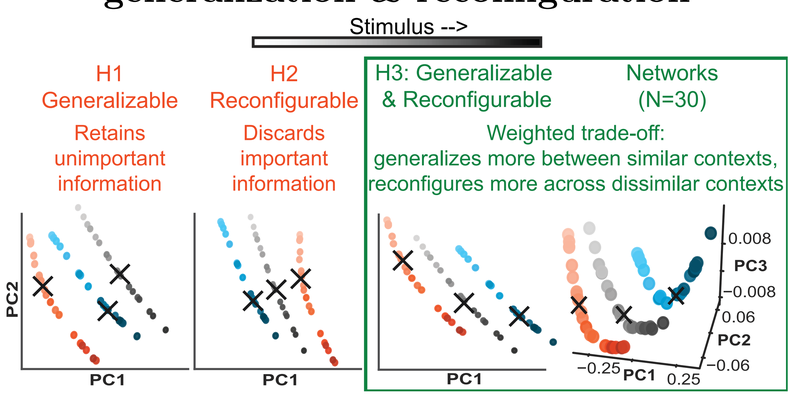
Behavioral hypotheses

A *generalizable* stimulus representation that *reconfigures* with context

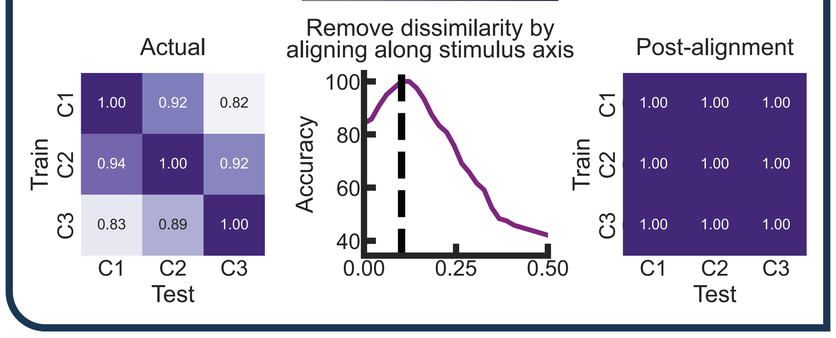


Network representations

Computational benefits of balancing generalization & reconfiguration



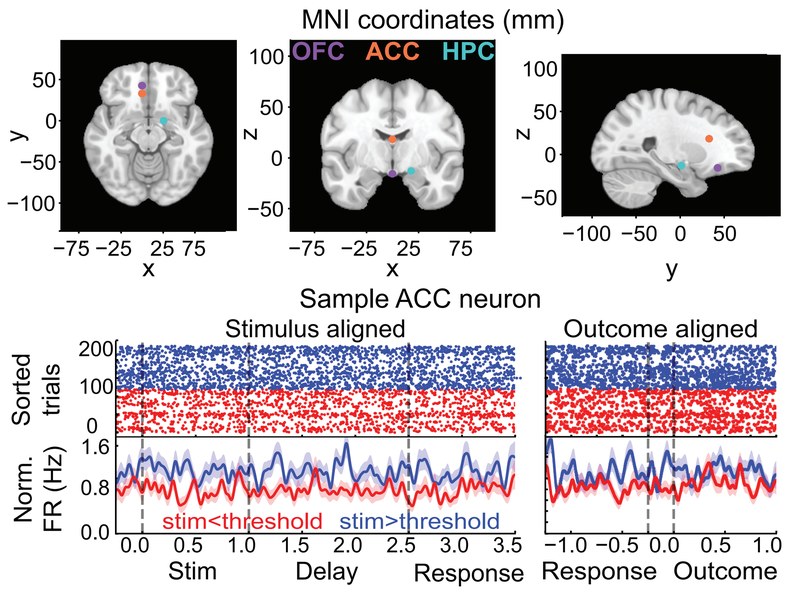
Reconfiguration scales with context dissimilarity



Acknowledgements

Columbia Engineering Startup Grant (N.R.), R01:MH128187 (E.H.S.), CCN volunteering committee membership (S.I.).

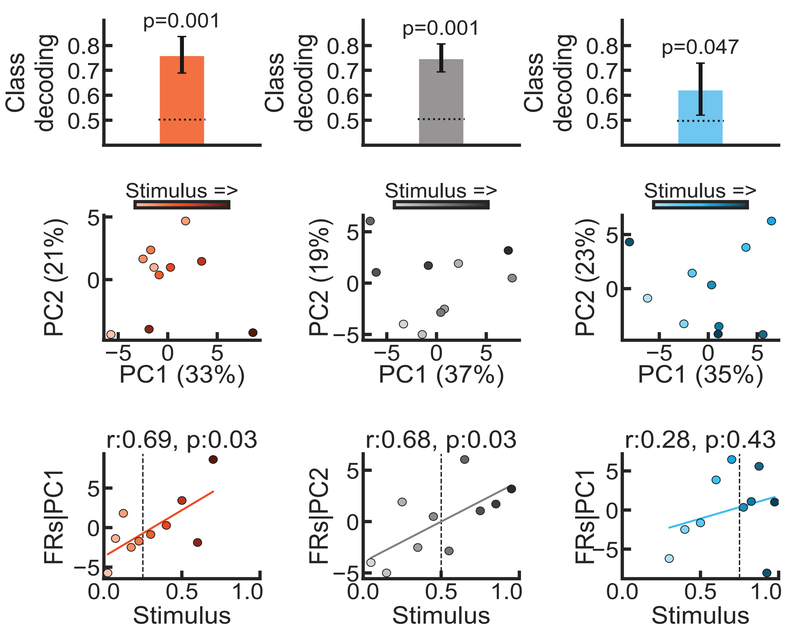
Human single-neuron data



22 HPC, 18 ACC, and 13 OFC neurons identified across 4 patients

Preliminary neural analyses (stimulus epoch)

Stable, linear representation of stimulus per context



Conclusion

- Human behavior & network simulations maintain generalizable context representations, whose reconfigurations scale with distance between contexts.
- Preliminary human PFC & HPC neuronal analyses show stable representations per context, which will next be tested on our simulated predictions.
- This builds on work suggesting that the PFC/HPC may form a context space or map, along which stable representations can be efficiently reconfigured.

References

- Courellis, H., Minxha, J., ... Fusi, S., Rutishauser, U., 2024. *Nature.*
- Henderson, M., Serences, J., and Rungratsameetaweemana, N., 2025. *Nature Comms.*
- Nogueira, R., Esteki, S., Fusi, S., Kiani, R., 2026. *bioRxiv.*